

Note: ECO2010 Problem Set 4, Question 2

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Ω UHC Argument

We want to show that $\Omega(\mathbf{p}, b)$ is UHC. Suffices to show that, given $(\mathbf{p}_0, b_0)$, with $b_0 > 0$ and $\mathbf{p}_{0,(i)} > 0 \forall i$, and V open such that $\Omega(\mathbf{p}_0, b_0) \subseteq V$, there exists $\delta > 0$ such that $\Omega(\mathbf{p}, b) \subseteq V$ for all $(\mathbf{p}, b) \in N_\delta(\mathbf{p}_0, b_0)$, because then the open set “ U ” needed to satisfy the UHC requirement can just be defined as $U := N_\delta(\mathbf{p}_0, b_0)$.

For convenience, we will use the max metric throughout, i.e.

$$d(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} - \mathbf{y}\|_\infty := \max_i |\mathbf{x}_{(i)} - \mathbf{y}_{(i)}|.$$

Since V is open and $\Omega(\mathbf{p}_0, b_0) \subseteq V$, we have that, for all $\mathbf{x} \in \Omega(\mathbf{p}_0, b_0)$, there exists $\epsilon_{\mathbf{x}} > 0$ such that $N_{\epsilon_{\mathbf{x}}}(\mathbf{x}) \subseteq V$. Now, $\{N_{\epsilon_{\mathbf{x}}}(\mathbf{x}) \mid \mathbf{x} \in \Omega(\mathbf{p}_0, b_0)\}$ is an open cover of $\Omega(\mathbf{p}_0, b_0)$ and therefore, since $\Omega(\mathbf{p}_0, b_0)$ is compact, this open cover has a finite subcovering, i.e. there exists a finite set $\{\mathbf{x}_1, \dots, \mathbf{x}_K\} \subseteq \Omega(\mathbf{p}_0, b_0)$ such that $\Omega(\mathbf{p}_0, b_0) \subseteq \bigcup_{k=1}^K N_{\epsilon_{\mathbf{x}_k}}(\mathbf{x}_k) \subseteq V$. Define

$$\epsilon := \min_{k \in \{1, \dots, K\}} \frac{\epsilon_{\mathbf{x}_k}}{2} > 0.$$

Now, it suffices to show that there exists $\delta > 0$ such that, for all $(\mathbf{x}, \mathbf{p}, b)$ such that $(\mathbf{p}, b) \in N_\delta(\mathbf{p}_0, b_0)$ and $\mathbf{x} \in \Omega(\mathbf{p}, b)$, there exists $\mathbf{x}^\dagger \in \Omega(\mathbf{p}_0, b_0)$ such that $\mathbf{x} \in N_\epsilon(\mathbf{x}^\dagger)$.

Now, define $\underline{p} := \min_i \mathbf{p}_{0,(i)} > 0$, and note that, for sufficiently small $\delta > 0$, we must have $\mathbf{p}_{\delta,(i)} \geq \underline{p}/2$, and so, we have

$$\mathbf{x}_{\delta,(i)} \leq \frac{b_\delta}{\mathbf{p}_{\delta,(i)}} \leq \frac{2b_\delta}{\underline{p}} < \frac{2(b_0 + \delta)}{\underline{p}},$$

which implies

$$\begin{aligned} \mathbf{p}_0 \cdot \mathbf{x}_\delta &= \mathbf{p}_\delta \cdot \mathbf{x}_\delta + (\mathbf{p}_0 - \mathbf{p}_\delta) \cdot \mathbf{x}_\delta \leq b_\delta + n\delta \|\mathbf{x}_\delta\|_\infty \leq b_\delta + n\delta \left(\frac{2(b_0 + \delta)}{\underline{p}} \right) \\ &\leq b_0 + \delta + n\delta \left(\frac{2(b_0 + \delta)}{\underline{p}} \right). \end{aligned}$$

Now, we have that

$$\mathbf{p}_0 \cdot (\lambda \mathbf{x}_\delta) \leq \lambda \left(b_0 + \delta + n\delta \left(\frac{2(b_0 + \delta)}{\underline{p}} \right) \right) ,$$

and so, setting $\mathbf{x}_\delta^\dagger := \lambda_\delta^\dagger \mathbf{x}_\delta$, where

$$\lambda_\delta^\dagger := \frac{b_0}{b_0 + \delta + n\delta \left(\frac{2(b_0 + \delta)}{\underline{p}} \right)} \in (0, 1) ,$$

and we have that

$$\mathbf{p}_0 \cdot \mathbf{x}_\delta^\dagger \leq b_0 .$$

Finally, note that $\lambda_\delta^\dagger$ approaches 1 as δ approaches 0, and so,

$$\|\mathbf{x}_\delta - \mathbf{x}_\delta^\dagger\|_\infty = \left\| (1 - \lambda_\delta^\dagger) \mathbf{x}_\delta \right\|_\infty \leq (1 - \lambda_\delta^\dagger) \frac{2(b_0 + \delta)}{\underline{p}}$$

can be made arbitrarily small by picking δ sufficiently small. The result now follows immediately by picking $\delta > 0$ small enough that

$$(1 - \lambda_\delta^\dagger) \frac{2(b_0 + \delta)}{\underline{p}} < \epsilon .$$